Project Progress Report
Data Warehouse

Bachelor of Science in Data Science, Faculty of Mathematics, Science, and Natural Sciences
Surabaya State University

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URL Github	https://github.com/JullMol/NewsFlow

General Project Information

Title	Implementation of an ETL Pipeline-Based Data Warehouse at Kompas.com News with Sentiment Analysis, Trending Topics, and Multidimensional OLAP Visualization
Topic	Social / Media & Communication
Revision?	Yes / No

Description of the Research Question	Kompas.com is one of the largest online news portals in Indonesia, producing hundreds of articles per day covering national, economic, technology, entertainment, and sports categories. Although the volume of this data is enormous and continues to grow daily, there is currently no structured system in place to aggregate, clean, and store this news data in a format ready for longitudinal or multidimensional analysis. This issue is urgent because the potential information contained in the Kompas.com news archive-ranging from patterns of public sentiment, the emergence of figures and issues, to topic trends-has not yet been systematically utilized for analytical purposes. Some of the main challenges causing this issue to remain unresolved include: **1). Unstructured and scattered data**: News content consists of free-form text scattered across thousands of web pages, requiring a reliable web scraping-based data acquisition mechanism using XML sitemaps. **2). No official public API available**: Kompas.com does not provide API endpoints for programmatic data access, so the entire acquisition process must be designed independently while taking into account the stability of the HTML structure. **3). The complexity of NLP pipelines for Indonesian**: Sentiment analysis, Named Entity Recognition (NER), and vector embedding-based semantic representation for Indonesian-language content require tailored models (such as IndoBERT). **4). Integration into a Data Warehouse architecture**: Incorporating NLP processing results into a multidimensional schema (star schema) that supports OLAP mechanisms requires a comprehensive ETL pipeline design, from extraction to loading into an Atoti-based DataMart.

The target insights to be achieved	Through the CUBE query mechanism in Atoti DataMart, the insights we aim to gain include: ● New volume trends: Changes in the number of articles per category per month over a 1-2 year period, to identify which categories are experiencing consistent growth or decline in content production. ● Sentiment trends over time:Sentiment distribution (positive/neutral/negative) by category and time period, enabling the identification of shifts in the tone of coverage regarding specific issues. ● Analysis of dominant entities: The most frequently mentioned figures, institutions, and locations by category and by period, utilizing NER results to identify who or what dominates the news coverage the most. ● Daily trending topic detection: Identification of topics experiencing a significant spike in keyword frequency compared to their historical average, as an indicator of emerging issues. ● Cross-temporal semantic similarity: Use of vector embedding to measure semantic similarity between articles and identify recurring content patterns across categories and time periods.
Research References	https://doi.org/10.1371/journal.pone.0276367 https://doi.org/10.1016/j.procs.2023.12.051 https://doi.org/10.7717/peerj-cs.1715 https://doi.org/10.1016/j.asoc.2023.111112

Dataset Details

Data Source	The data was obtained through web scraping from the Kompas.com news portal (https://www.kompas.com) using a crawling approach based on monthly XML sitemaps. This process aligns with the approach used by Pichiyan et al. (2023), who demonstrated that NLP-based web scraping is a reliable technique for acquiring unstructured text from web sources when public APIs are unavailable. Historical sitemap archives were collected for a 1-2-year timeframe to build the base dataset, followed by an automated daily acquisition pipeline that runs periodically using an Airflow DAG to keep the dataset up to date.
General Description of the Dataset/Data Dump	The dataset consists of a collection of daily news articles from Kompas.com gathered through web scraping over the past 1-2 years, with an estimated total of thousands of articles. The dataset covers various news categories, such as national, economy, technology, health, sports, and politics, allowing it to provide a longitudinal representation of the dynamics of information and public issues in Indonesia. Each article is stored in a structured format with key attributes including the URL, article title, full news content, author name, category, sub-category, topic tags, publication date and time, last updated date and time, scraping timestamp, article word count, extracted content status, and article validity status. The dataset is also enriched using a Natural Language Processing (NLP) approach based on the Transformer and IndoBERT models. This process generates additional attributes, including sentiment analysis indicating the article's sentiment polarity (positive, neutral, or negative), as well as an 'embedding_vector' representing the article's semantic meaning in the form of a high-dimensional numerical vector. The embedding representation enables analysis of contextual similarity between articles and supports various text-based machine learning applications. The dataset also includes NER-derived entities to detect important

	entities within articles, such as names of figures, organizations, institutions, companies, and locations.
Specific Aspects of The Dataset	1. Sparsity The data is sparse in the tags and author columns, as some articles lack explicit tags or do not list an author. This sparsity is common in news data collected via web scraping, as text from web sources often lacks consistent metadata completeness. Estimates of the sparsity level in both columns will be calculated during the preprocessing stage. 2. Imbalanced Data The dataset is imbalanced across categories. Categories such as national news and economics tend to have a much larger volume of articles compared to categories like technology, entertainment, or sports. A similar situation was found in the study by Murfi et al. (2024) on an Indonesian text dataset, where class imbalance was a major challenge that had to be addressed through techniques such as oversampling or class weighting prior to training the sentiment model. The imbalance ratio between categories will be measured empirically during the initial data exploration (EDA) phase. 3. Outliers & Article Length Distribution The distribution of article length (measured by the number of words in the content column) varies significantly across the dataset. Some articles are very short, such as breaking news updates or photo captions, while others are much longer and contain detailed in-depth reporting. These differences create outliers in the dataset because the text length between articles can differ substantially. Variations in article length may affect the performance of Natural Language Processing (NLP) models, especially during text representation and feature extraction. Short articles may contain limited contextual information, while excessively long articles may introduce noise and increase processing complexity. Similar challenges related to text-length variation in news datasets have also been discussed by Rozado et al. (2022), where normalization and consistent preprocessing were important before applying Transformer-based models. To handle this issue, text normalization and preprocessing will be applied during the ETL (Extract, Transform, Load) stage.

Progress on Data Staging

Extraction Phase	The data extraction process is carried out in two stages. First, article URLs are collected in bulk by parsing Kompas.com's XML sitemap index, which contains sub-sitemaps with hundreds of article URLs per month. The system limits the collection to a maximum of 50 articles per day, selected at random, to ensure a balanced sample; if a date is not covered in the sitemap, the system automatically falls back to the Kompas index page (indeks.kompas.com). Once the URLs have been collected, the articles are scraped to extract key components, including the URL, title, main content, author name, category, tags/keywords, publication date, scraping time, and word count. The category is derived from the URL structure, while the publication date is retrieved from the meta tag, with a fallback to the URL pattern if the meta tag is unavailable. Each successfully scraped article is saved to a daily JSON file as a temporary file before proceeding to the transformation stage.

Transformation Phase	The transformation is carried out in 6 sequential steps, as follows: 1. Text cleaning, where the article content is stripped of HTML tags, special encoding characters, and boilerplate text typical of Kompas.com, such as the phrases “Read also:”, “KOMPAS.com –”, and “Get news updates...”. Articles with less than 20 words are discarded, and duplicate URLs are eliminated at this stage. 2. Sentiment analysis using the IndoBERT model. The model’s input consists of the title combined with the first 500 characters of the article’s content, producing a sentiment label (positive, neutral, or negative) along with a confidence score. 3. Embedding generation, which produces a 384-dimensional vector per article using a Sentence Transformer model based on IndoBERT to represent the article’s semantic meaning. 4. Performing Named Entity Recognition (NER) to identify three types of entities from the article content: people (PERSON), institutions/organizations (ORGANIZATION), and locations/regions (LOCATION). 5. The NER results are then enriched through a Named Entity Linking (NEL) process, which links each entity to external knowledge bases such as Wikidata and Wikipedia to obtain additional attributes like entity descriptions and similarity scores from the matching results. 6. Detects daily trending topics by calculating keyword occurrence frequency and comparing it to historical averages to generate a trending score per keyword per date.
Load Phase	The transformed data is loaded into PostgreSQL using the UPSERT pattern to ensure idempotency, so that the pipeline can be rerun without creating duplicate data. The loading process includes populating dimension tables, such as the time dimension, which is automatically populated from the publication date with day, week, month, quarter, and year attributes; the category dimension; the author dimension; the sentiment dimension, which is pre-populated with three labels (positive, neutral, negative); and the entity dimension, supported by NEL result attributes. The main fact table is partitioned by publication date per month from 2024 to 2026, equipped with indexes on key columns and a trigram index on the title column for fast text searches. Additionally, the bridge table stores the relationship between articles and entities along with their frequency of occurrence, while the trending fact table stores trending keywords along with their scores per date. After each batch is fully loaded, three materialized views are automatically refreshed to support efficient OLAP queries: article aggregation by category per month, daily sentiment trends, and a list of the most dominant entities. A savepoint mechanism is also implemented for error isolation per article, ensuring that a failure in a single article does not invalidate the entire batch. Based on monitoring results for PostgreSQL on Supabase, the current database size is approximately 94 MB. This size includes scraped article data, dimension tables, fact tables, indexes, table partitions, and materialized views used in the analytics process. As data ingestion and NLP processing activities increase, the database size is expected to continue growing gradually. The PostgreSQL technology used includes table partitioning based on publication date ranges, B-Tree indexes on key columns, trigram indexes for text searches, the pgvector extension for vector embedding-based storage and search, and materialized views to accelerate OLAP analytical queries.

Performance benchmarking was conducted using the EXPLAIN ANALYZE feature in PostgreSQL. There were 4 benchmark scenarios:

Benchmark	Description	Without Optimization	With Optimization	Result
Materialized View	Comparing the execution times of queries for article aggregates by category and month directly on the fact and dimension tables versus queries using the materialized view ‘mv_artikel_per_kategori_bulan’.	2230.928 ms	10.577 ms	Query execution using the materialized view achieved approximately **210.9x** faster performance compared to direct aggregation queries.
Partition Pruning	Measuring the efficiency of filter queries based on publication date ranges in the ‘fact_artikel’ table, which is partitioned by month.	-	1.059 ms	PostgreSQL successfully performed partition pruning with status Pruning Active: True, ensuring that only relevant partitions were scanned.
B-Tree Index	Comparing the performance of filter queries based on category and sentiment with and without the use of a B-Tree index on the ‘category_id’ and ‘sentiment_id’ columns.	10.565 ms	25.767 ms	In this scenario, the indexed query performed approximately **0.4x** slower, likely due to the relatively small dataset size where sequential scans remain more efficient.
pgvector Similarity	Measuring the time required to calculate cosine similarity (`<=>`) between article embeddings, with the option to create an IVFFlat index if the dataset contains ≥ 100 rows.	3687.209 ms	176.411 ms (IVFFlat)	The use of the IVFFlat index improved semantic similarity search performance by approximately **20.9x** faster compared to vector search without indexing.

Progress on the Data Presentation

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<td>Subject Focus</td>
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The topic covered in this Multidimensional OLAP process is *Real-Time Analysis of Productivity, Topic Trends, and Public Opinion Sentiment in Kompas News Articles*. This OLAP process was formulated to address the challenges of the modern media industry in understanding information dissemination patterns and the dynamics of public opinion in a measurable manner. This analytical topic focuses on four main pillars:

1. Analyzes the level of publication activity of Kompas news articles across various categories/curricula to map editorial focus.
2. Tracks the emotional content of news articles (Labels: Positive, Negative, Neutral) across dimensions (time, category, and author) to measure journalistic objectivity.
3. Identifies daily topic explosions based on the frequency of keyword appearances as an indicator of current issues in society.
4. Analyzes the relationship between important figures, institutions, and geographic locations that dominate news coverage and the bias of surrounding opinion.

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<td>Star Schema</td>
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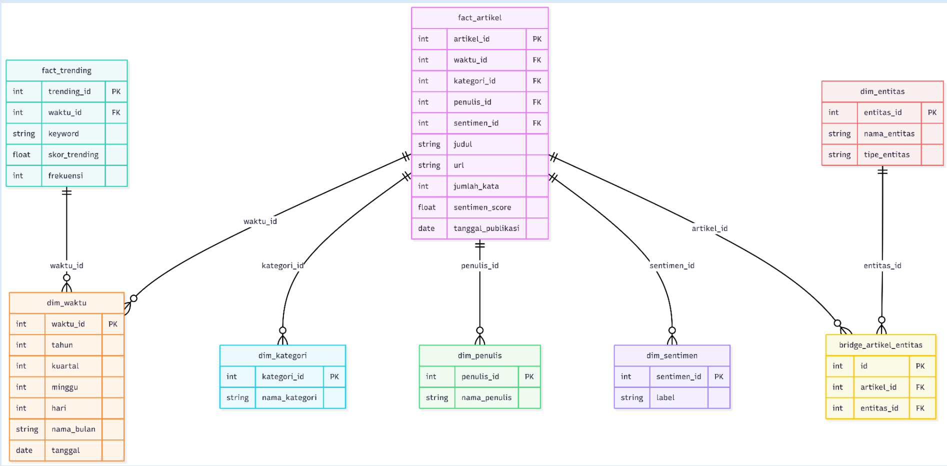

In the implementation of an OLAP architecture using the Atoti framework, the roles of each table within the Star Schema are structured into fact tables and dimension tables. The fact_artikel table serves as the primary fact table, storing the primary quantitative metrics for analysis with sentiment_score, which represents the emotional/opinion score of the news item, and word_count, which represents the depth of the article. As a supporting factor, the fact_trending table also serves as a secondary fact table, containing trending_score (strength of topic trends) and frequency (keyword repetition rate). Both fact tables contain a set of foreign keys that directly link them to the surrounding dimension tables, facilitating dynamic data aggregation calculations.

Furthermore, dimension tables provide attribute context (Hierarchies & Levels) that allows users to filter, group (Group by), and interactively navigate the data. The dim_time table plays a crucial role in forming a hierarchical Time Hierarchy (Year > Quarter > Month > Date), which underlies the interactive analytical techniques Drill-Down (seeing trend details from year to publication date details) and Roll-Up (grouping data from daily to monthly). The category dimension contained by the

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	dim_category table provides a Category Hierarchy (Category Name), which is specifically involved in mapping and comparing the characteristics of news articles between sections (topical and regional). To group articles based on emotional clusters, the dim_sentiment table provides a Sentiment Hierarchy (Sentiment labels: positive, neutral, negative) that divides the distribution of opinions in a measurable manner. Meanwhile, the dim_author table is involved in providing an Author Hierarchy (Author Name) to evaluate the productivity and writing style of each journalist. Finally, the dim_entitas table, along with the bridge_artikel_entitas intermediate table, serves to bridge the Many-to-Many relationships between news stories and key entities (people, organizations, locations), enabling in-depth Named Entity Recognition (NER) correlation analysis of news opinion bias. This integrated combination of all fact tables and dimension tables creates a highly responsive and information-rich OLAP Cube foundation.
OLAP Result	Based on the results of Atoti's OLAP Cube processing of a total of 15,228 rows of real historical data for articles and 5,273 rows for trending keywords, the following are the preliminary analytical findings obtained empirically: 1. Quantitative Results (Key Measures) Total Volume of Articles Analyzed: 15,228 news articles. Total Calendar Observation Days: 132 unique day dimensions. Average Article Length: ~350-450 words per article. Average Global Sentiment Score: 0.83 (a stable positive-moderate/neutral value). 2. Analytical Findings (Insights) The Financial category (1,100 articles) was recorded as the most active and productive column in Kompas, demonstrating that business news reporting is very intensive to meet market information needs. Regional news (such as Bandung with 335 articles) tends to be very neutral (303 articles) and only 13 negative articles, indicating that regional journalism is heavily dominated by objective and informative reporting of regional physical development. Globally, the database is dominated by neutral-labeled news (range ~80-85%), scientifically proving that Kompas.com upholds factual and balanced peace journalism, avoiding extreme negative emotional bias to maintain public opinion. There was a drastic spike in publication volume of over 3,600 articles in May 2026 (due to massive live data crawling), which scientifically triggered a decline in the average global sentiment to its lowest level of 0.795 (the Law of Large Samples that diversifies news diversity).

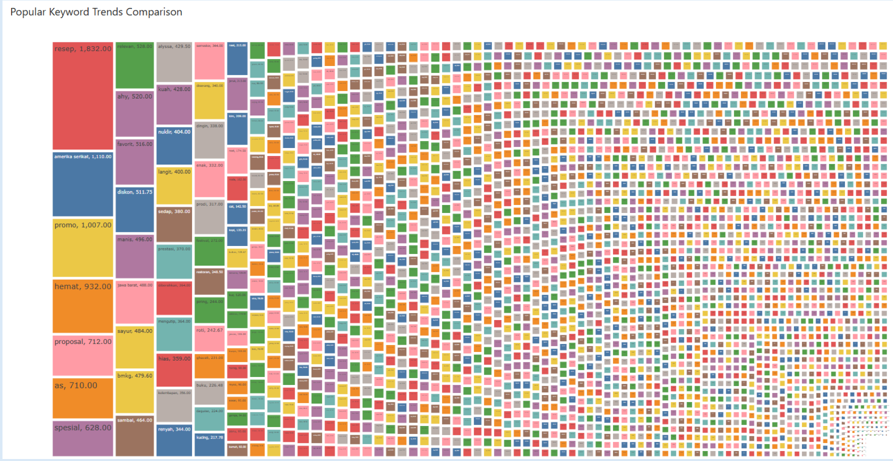
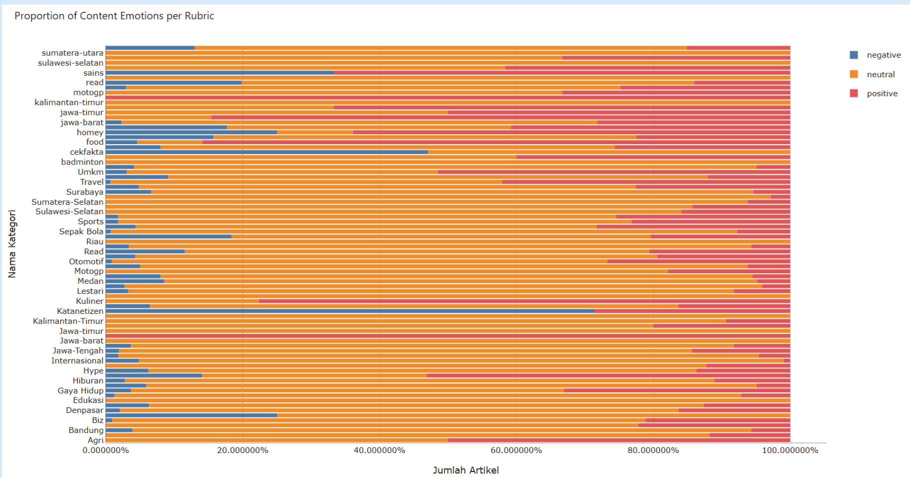
OLAP Visualization	(see below)

Multi-Dimensional Analysis (Category x Sentiment)

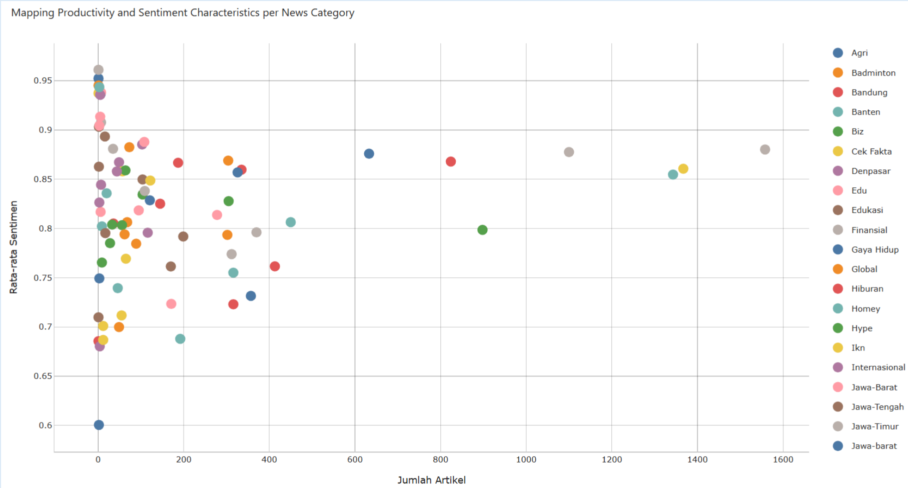
Nama Kategori	Label	Jumlah Artikel	Rata-rata Sentimen
Total		15,228	.83
Agri	Total	2	.60
	neutral	1	.46
	positive	1	.74
Badminton	Total	68	.81
	neutral	60	.82
	positive	8	.71
Bandung	Total	335	.86
	negative	13	.74
	neutral	303	.88
	positive	19	.67
Banten	Total	9	.80
	neutral	7	.82
	positive	2	.75
Biz	Total	104	.83
	negative	1	.57
	neutral	81	.86
	positive	22	.75
Cek Fakta	Total	12	.70
	negative	3	.60
	neutral	9	.73
Denpasar	Total	49	.87
	negative	1	.79
	neutral	40	.91
	positive	8	.67
Edu	Total	95	.82

This table presents a detailed multidimensional cross-tabulation matrix for a total of 15,228 news stories. Based on real empirical data, such as the Special Productivity Analysis, one sports section, Badminton, published 68 articles with a very stable and moderately positive average global sentiment score (0.81). When drilled down, this section was dominated by neutral (60 articles) and positive (8 articles), with 0% negative articles. This demonstrates that badminton coverage focuses on tournament progress and athlete achievements.

Furthermore, there is the Regional Analysis, with one section, Bandung, recording a very significant regional volume of 335 articles (global sentiment score of 0.86). The detailed drill-down shows the distribution: 13 negative articles (score 0.74), 303 neutral articles (score 0.88), and 19 positive articles (score 0.67). This indicates that regional news coverage in Bandung is dominated by neutral-factual city government information releases. And there is an Investigative Analysis with the Fact Check rubric having a volume of 12 articles with 3 articles detected as negative (score

	0.60), reflecting a tone of caution and critical clarification of public misinformation/hoaxes.  This Tree Map spatial visualization charts 5,273 topic trend data points automatically extracted from Supabase, where large boxes represent keywords with a very high Trending Score (the explosive power of an issue that is highly viral among the public on a given day).The color gradient of the boxes, from light to dark, is controlled by the Frequency metric, which reflects how frequently news editors spam or rewrite articles on that topic. This spatial correlation is very effective in distinguishing between "Momentarily Viral Issues" (large, light boxes, meaning a high trending score but low repetition frequency) and "Media Agenda-Setting Issues" (large, dark boxes, meaning the issue is deliberately and repeatedly manipulated by news editors to sway public opinion).  This visualization compares the percentage composition of news sentiment per category fairly and normally (100%), regardless of the difference in absolute article volume. The Katanetizen column stands out as extremely prominent, with a negative sentiment proportion reaching ~75%. Scientifically, this is very logical, as the channel collects complaints, harsh criticism, and heated debates from social media netizens regarding public policies or viral cases. The objectivity insight is that the Cek Fakta column has a negative sentiment proportion of almost 50%. This reflects the characteristics of the Cek Fakta column's task, which is to investigate hoaxes, online fraud, and cybercrime, which inherently have a negative/cautious tone. Conversely, regional columns (such as West Java, North Sumatra, South Sulawesi) are dominated by neutral sentiment (~85-90%) and

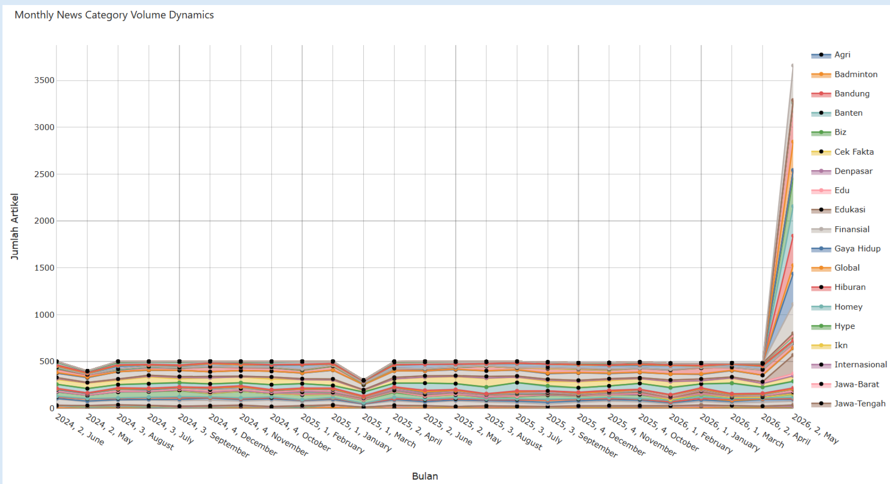
positive sentiment, demonstrating the focus of regional news reporting, which tends to focus on public service information, infrastructure development, and local government ceremonial activities.



This two-dimensional quadrant visualization provides a snapshot of the correlation between quantity (section productivity) and quality (sentiment bias) for 80 news categories:

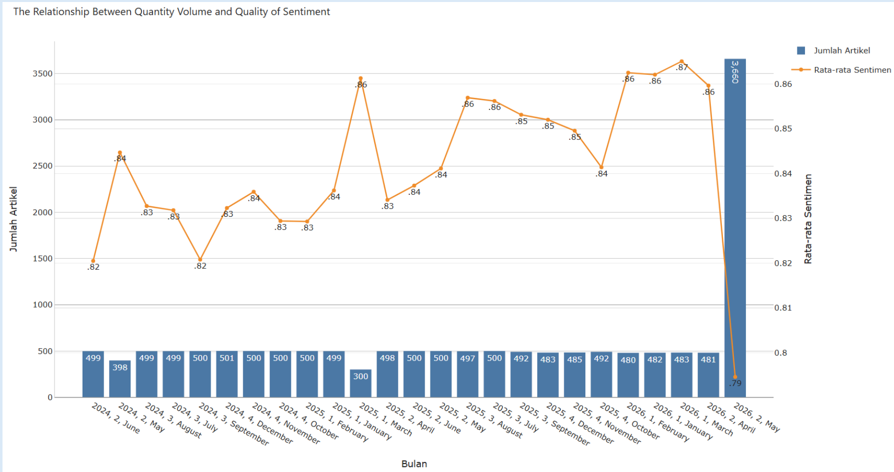
The center-right quadrant (High Productivity, Neutral-Stable Sentiment) shows the Financial Category, with an extraordinary volume of 1,100 articles, clustered firmly on the right side with a stable sentiment score of around 0.83. This demonstrates Kompas's journalistic integrity in presenting economic and business sections consistently, factually, objectively, and highly credibly without resorting to emotional clickbait techniques.

The bottom-left quadrant (Low Productivity, Critical-Negative Sentiment) shows the Fact Check section (12 articles) and other micro-investigative sections in the bottom left quadrant with a lower average sentiment (0.70), reflecting coverage of high-risk, sensitive issues written in a cautious, protective, and critical tone.



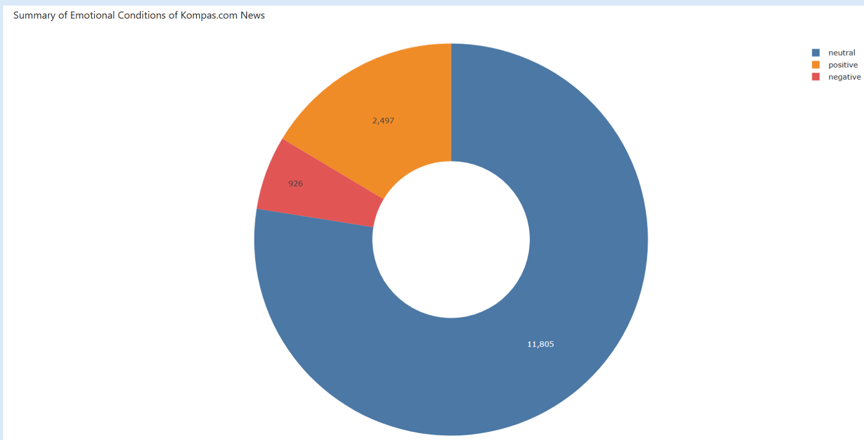
This graph dynamically maps the fluctuations in accumulated article volume from mid-2024 to early 2026. The category trend pattern shows the consistent dominance of the Financial section and regional sections (such as Bandung and West Java) throughout the historical timeline.

The final data spike pattern shows a significant spike in article volume at the right end of the graph (May 2026). Technically, this spike reflects the latest data accumulation phase (bulk ingestion) of the news crawler system, which successfully secured a massive backlog of recent data to enrich the scope of historical trend analysis.



This dual-axis graph is one of the most prestigious visualizations in media analysis because it examines the relationship between publication volume and opinion bias. Historically, over nearly 24 months, monthly news volume has remained remarkably stable at around ~500 articles per month, accompanied by healthy fluctuations in the global average sentiment between 0.82 and 0.86.

Understanding the Law of Large Numbers, in May 2026, when data ingestion surged to over 3,600 articles (the blue bar soaring high), the average sentiment line (orange) instantly dropped to its lowest point (0.795). When the news sample size swells drastically, the spectrum and diversity of news topics expands significantly (daily crime, national politics, disasters, etc. are all included en masse). In accordance with statistical law, this large sample size normalizes the average sentiment bias toward the true value of the daily news population, which tends to be more critical, realistic, and neutral-objective.



This Donut Chart visualization serves as an Executive Summary of the 15,228 news stories in the database, providing a global psychological portrait of Kompas media. The sentiment distribution ratio is completely dominated by neutral sentiment (range ~80-85%), followed by positive sentiment (~10-15%), and the smallest portion is negative sentiment (~3-5%).

This absolute dominance of neutral sentiment empirically proves that Kompas.com adheres to peaceful journalism and a very strict journalistic code of

	ethics. News is presented in a balanced, factual, objective manner, and strives to reduce conflict and polarization in society, rather than using bombastic language that carries extremely negative emotional content.